

Benjamin Boe

Wentworth Institute of Technology
School of Sciences & Humanities
550 Huntington Avenue
Boston, MA 02115

Email: boeb@wit.edu

Education

Ph.D. , Astronomy, Institute for Astronomy, University of Hawai'i (IfA/UH) Advisor: Shadia Habbal Thesis: "Solar Wind Formation in the Corona"	August 2020
M.S. , Astronomy, IfA/UH	May 2018
B.S. , Physics and Mathematics, University of Puget Sound (UPS) Cum laude and Honors in Physics	May 2015

Professional Appointments

Assistant Professor , Wentworth Institute of Technology (WIT)	August 2023-present
NSF Postdoctoral Fellow , IfA/UH	October 2020–March 2023
Faculty Lecturer , Physics & Astronomy, UH	August 2021–December 2021

Observing Experience

Total Solar Eclipse Expeditions	
Preparation in lab and on-site operation, alignment, and calibration of instrumentation	
Texas, USA	April 8th, 2024
Exmouth, Australia	April 20th, 2023
Antarctic Ocean	December 4th, 2021
Vicuña, Chile	July 2nd, 2019
Nebraska, USA	August 21st, 2017
Halmahera, Indonesia	March 9th, 2016
Maunakea Observing	
Subaru HSC, summit observing	January 3rd and 4th, 2016

Professional Development

Referee	
- The Astrophysical Journal, 9 articles	2020–2025
- Solar Physics, 4 articles	2021–2025

Publications

Peer-Reviewed Articles

Solar Physics

- Benavitz, L. F., **Boe, B.**, Habbal, S. R. *Total Solar Eclipse White-light Images as a Benchmark for Potential Field Source Surface Coronal Magnetic Field Models: An In-depth Analysis over a Solar Cycle*. ApJ, 974, 178, 2024. <https://iopscience.iop.org/article/10.3847/1538-4357/ad71c6>
- Zhu, Y., Habbal, S.R., Ding, A., Yamashiro, B., Landi, E., **Boe, B.**, Constantinou, S., and Nassir, M., *Spectroscopic Observations of the Solar Corona during the 2017 August 21 Total*

Solar Eclipse: Comparison of Spectral Line Widths and Doppler Shifts between Open and Closed Magnetic Structures, ApJ, 699, 122, 2024. <https://iopscience.iop.org/article/10.3847/1538-4357/ad3424>

3. **Boe, B.**, Downs, C., Habbal, S., *The Solar Minimum Eclipse of 2019 July 2. III. Inferring the Coronal T_e with a Radiative Differential Emission Measure Inversion*, ApJ, 951, 55, 2023. <https://iopscience.iop.org/article/10.3847/1538-4357/acd10b>
4. **Boe, B.**, Habbal, S., Downs, C., Druckmüller, M., *The Solar Minimum Eclipse of 2019 July 2. II. The First Absolute Brightness Measurements and MHD Model Predictions of Fe X, XI, and XIV out to $3.4 R_\odot$* , ApJ, 935, 173, 2022. <https://iopscience.iop.org/article/10.3847/1538-4357/ac8101>
5. **Boe, B.**, Yamashiro, B., Druckmüller, M., and Habbal, S., *The Double-bubble Coronal Mass Ejection of the 2020 December 14 Total Solar Eclipse*, ApJ Letters, 914, L39, 2021. <https://iopscience.iop.org/article/10.3847/2041-8213/ac05ca>
6. Habbal, S., Druckmüller, M., Alzate, N., ... **Boe, B.**, et al. *Identifying the Coronal Source Regions of Solar Wind Streams from Total Solar Eclipse Observations and in situ Measurements Extending Over a Solar Cycle*, ApJ Letters, 911, L4, 2021. <https://iopscience.iop.org/article/10.3847/2041-8213/abe775>
7. **Boe, B.**, Habbal, S., Downs, C., Druckmüller, M., *The Color and Brightness of the F-Corona Inferred from the 2019 July 2 Total Solar Eclipse*, ApJ, 912, 44, 2021. <https://iopscience.iop.org/article/10.3847/1538-4357/abea79>
8. **Boe, B.**, Habbal, S., Druckmüller, M., *Coronal Magnetic Field Topology From Total Solar Eclipse Observations*, ApJ, 895, 123, 2020. <https://doi.org/10.3847/1538-4357/ab8ae6>
9. **Boe, B.**, Habbal, S., Druckmüller, M., Ding, A., Hodérova, J., and Štarha, P., *CME-induced Thermodynamic Changes in the Corona as Inferred from Fe XI and Fe XIV Emission Observations during the 2017 August 21 Total Solar Eclipse*, ApJ, 888, 100, 2020. <https://doi.org/10.3847/1538-4357/ab5e34>
10. **Boe, B.**, Habbal, S., Druckmüller, M., Landi, E., Kourkchi, E., Ding, A., Štarha, P., & Hutton, J., *The First Empirical Determination of the Fe^{10+} and Fe^{13+} Freeze-in Distances in the Solar Corona*, ApJ, 859, 155, 2018. <https://doi.org/10.3847/1538-4357/aabfb7>

Long Period Comets

1. **Boe, B.**, Jedicke, R., Meech, K., Wiegert, P., Weryk, R., et al., *The orbit and size-frequency distribution of long period comets observed by Pan-STARRS1*, Icarus, 333, 252, 2019. <https://doi.org/10.1016/j.icarus.2019.05.034>

Galaxy Evolution

1. Shields, D., **Boe, B.**, Pfountz, C., Davis, B., et al., *Spirality: A novel way to measure spiral arm pitch angle*, Galaxies, 10, 100, 2022. <https://doi.org/10.3390/galaxies10050100>

Musical Acoustics

1. **Boe, B.**, and Randy Worland, *Experimental and numerical analysis of the effect of length in musical drumhead coupling*, Proceedings of Meetings on Acoustics, 20, 1, 2014. <http://asa.scitation.org/doi/abs/10.1121/1.4884779>

Non-Peer-Reviewed

White Papers for the 2024 Solar and Space Physics (Heliophysics) Decadal Survey

1. **Boe, B.**, Habbal, S., and Ding, A., *The Need for a New Generation of Space-based Visible and Near-IR Emission Line Observations of the Corona*, 2022, [Link](#)
2. Habbal, S., **Boe, B.**, Haggerty, C., and Ding, A., *A Total Solar Eclipse Earth-Based Mission: Multi-wavelength Observations from Land, Sea and Air to Probe the Critical middle Corona*, 2022, [Link](#)
3. Cooper, J., Habbal, S., **Boe, B.**, et al., *Heliophysics from, by, and of the Moon in the Age of Artemis: Lunar Solar Occultation Explorer (LunaSOX)*, 2022, [Link](#)

Talks and Presentations

Colloquia

1. “The Scientific Value of Total Solar Eclipses”, Invited Colloquium. California State University, Chico, CA, 2022
2. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Invited Colloquium. University of Puget Sound, Tacoma, WA, 2020
3. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Invited Colloquium. Bridgewater State University, Bridgewater MA, USA, 2020
4. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Invited Colloquium. Aberystwyth University, Aberystwyth, Wales, 2019
5. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Colloquium. Institut d'Astrophysique Spatiale, Orsay, France, 2019
6. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Colloquium. Brno University of Technology, Brno, Czech Republic, 2019
7. “Studying Solar Coronal Physics with Total Solar Eclipses”, Invited Colloquium. University of Puget Sound, Tacoma, WA, 2019
8. “Testing Solar System Formation With Manx Comets”, Colloquium. UH NASA Astrobiology Institute Seminar, University of Hawai'i, 2016

Conference Talks and Posters

9. “Advances in Total Solar Eclipse Narrowband Observations: Introducing a Radiative Differential Emission Measure”, Talk. American Geophysical Union (AGU) Meeting, District of Columbia, 2024
10. “Insights from a Total Solar Eclipse Research Expedition: Educating Students and Engaging the Public”, Talk. American Association of Physics Teachers (AAPT) meeting, Boston, MA, 2024
11. “The First Absolute Brightness Measurements and MHD Model Predictions of Fe X, XI, and XIV out to 3.4 Rs”, Talk. Triennial Earth-Sun Summit, Bellevue, WA, 2022

12. “The First Absolute Brightness Measurements and MHD Model Predictions of Fe X, XI, and XIV out to 3.4 Rs”, Poster. Solar Heliospheric and Interplanetary Environment (SHINE) Workshop, Honolulu, HI, 2022
13. “The Double-bubble Coronal Mass Ejection of the 2020 December 14 Total Solar Eclipse”, Poster. AGU Meeting, hybrid (virtual), 2021
14. “The Double-bubble Coronal Mass Ejection of the 2020 December 14 Total Solar Eclipse”, Poster. SHINE Workshop, virtual, 2021
15. “Coronal Magnetic Field Topology From Total Solar Eclipse Observations”, iPoster. American Astronomical Society (AAS) Solar Physics Division Meeting, virtual, 2020
16. “Thermodynamic Changes in the Corona during the 2017 August 21 Total Solar Eclipse”, Talk. AAS Meeting, Honolulu, HI, 2020
17. “CME Induced Thermodynamic Changes in the Corona as Inferred from Fe XI and Fe XIV Emission Observations during the 2017 August 21 Total Solar Eclipse”, Poster. AGU Meeting, San Francisco, CA, 2019
18. “The Orbit and Size-Frequency Distribution of Long Period Comets Observed by PS1”, Talk. EPSC-DPS Joint Meeting, Geneva, Switzerland, 2019
19. “CME Induced Thermodynamic Changes in the Corona as Inferred by Fe XI and Fe XIV Emission Observations from the 2017 August 21 Total Solar Eclipse”, Poster. SHINE Workshop, Boulder, CO, 2019
20. “First Empirical Determination of the Fe 10+ and Fe 13+ Freeze-in Distances in the Solar Corona”, Poster. SHINE Workshop, Cocoa Beach, FL, 2018
21. “First Empirical Determination of the Fe 10+ and Fe 13+ Freeze-in Distances in the Solar Corona”, Talk. Asia Oceania Geosciences Society Meeting, Honolulu, HI, 2018
22. “Breaking the Wall of Solar Wind Formation”, Talk. Falling Walls Lab, UH, 2017
23. “Testing Solar System Formation Models Using Pan-STARRS1 Detections of Nearly Inactive Manx Comets”, Talk. Division of Planetary Sciences Meeting, Pasadena, CA, 2016
24. “Pitch Angle Survey of GOODS Spiral Galaxies”, Poster. AAS Meeting, Seattle, WA, 2015
25. “Pitch Angle Survey of GOODS Spiral Galaxies”, Poster. Murdock College Science Research Program (MCSR) Conference, Vancouver, WA, 2014
26. “Experimental and numerical analysis of the effect of length in musical drumhead coupling”, Poster. Acoustical Society of America Meeting, San Francisco, CA, 2013
27. “Coupled Vibrations Between Musical Drumheads”, Talk. MCSR Conference, Vancouver, WA, 2013

Honors, Awards, and Grants

NSF SHINE Grant	\$385,959	2025-2028
NSF AGS Postdoctoral Research Fellowship	\$190,000	2020–2023
NSO Graduate Fellowship (AURA grant N97991C)	\$122,099	2018–2020
AAS Division of Planetary Science Travel Grant	\$650	2016
Sigma Pi Sigma Physics Honor Society		2015
University of Puget Sound Dean’s list		2014
Mortar Board Honor Society		2014
Raymond & Olive Seward Scholarship	\$3,500	2014

NSF Research Experience for Undergraduates grant	\$5,000	2014
Acoustical Society of America poster award	\$500	2013
John Van Zytveld Award in Physical Science	\$2,000	2013
Washington State NASA Space Grant	\$2,250	2013
Thomas & Hilda Jack Scholarship	\$2,750	2013
Puget Sound President's Scholarship	\$16,000/year	2011–2015
Kilworth Foundation Scholarship	\$2,500	2011